

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/17]

The following observations are recorded:

Intensity of unpolarised light = X

Intensity of polarised light after passing through first polaroid = Y

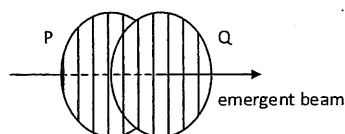
Intensity of polarised light after passing through second polaroid whose polarising axis is 30° to the first polaroid = Z

Which row give the correct intensity of X , Y and Z ?

	X	Y	Z
A	I_0	$\frac{1}{2}I_0$	$0.75I_0$
B	I_0	I_0	$0.75I_0$
C	$2I_0$	I_0	$0.75I_0$
D	$2I_0$	$\frac{1}{2}I_0$	$0.375I_0$

2. [AJC/H2/Prelims 2018/P1/14]

Two sheets of polaroid, P and Q, are placed so that their polarizing directions are parallel and vertical, as shown in the diagram. The amplitude of the emergent beam is A_0 .



What is the smallest angle through which Q must be turned for the amplitude of the emergent beam to be reduced to $\frac{1}{2}A_0$?

- A 30° B 45° C 60° D 90°

3. [CJC/H2/Prelims 2018/P1/13]

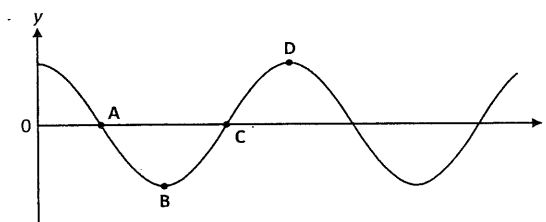
The phase difference between 2 points at a distance 60 cm apart along a progressive transverse wave is $\frac{1}{2}\pi$ rad.

If the frequency of the wave is 200 Hz, what is the speed of the wave?

- A 240 m s^{-1} B 480 m s^{-1} C 24000 m s^{-1} D 48000 m s^{-1}

4. [CJC/H2/Prelims 2018/P1/14]

A sound wave travelling towards the right through air causes the air molecules to be displaced from their original positions. The graph below shows the variation with distance x of the displacement y of the air molecules at a particular instant in time.



Taking the displacement towards the right as positive, at which point is the pressure maximum?

5. [DHS/H2/Prelims 2018/P1/16]

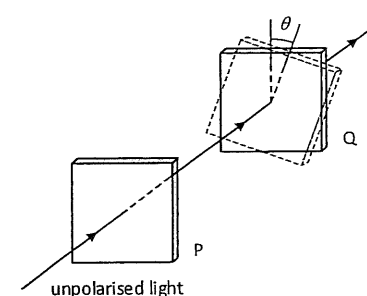
A beam of plane-polarised light of intensity I falls normally on a thin sheet of polaroid.

If the transmitted beam has an intensity of $\frac{1}{4}I$, what is the angle between the plane of incident polarisation and the polarising direction of the polaroid.

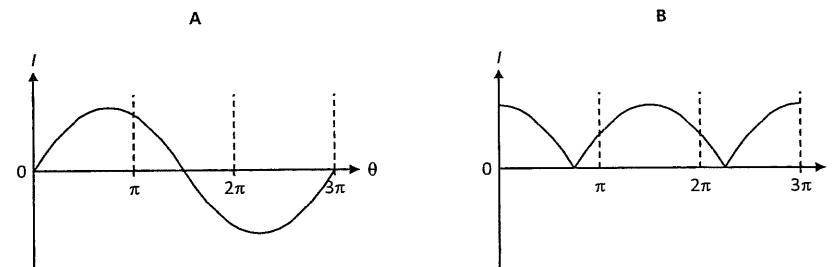
- A 22.5° B 30° C 45° D 60°

6. [EJC/H2/Prelims 2018/P1/15]

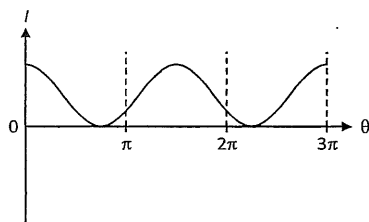
Unpolarised light is incident normally on a linear polarizer P and subsequently on a second linear polarizer Q as shown below.



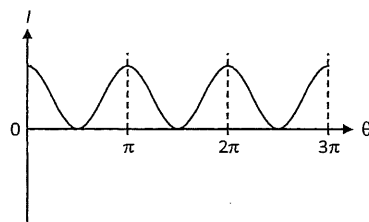
Which graph best represents the relationship between the emergent light intensity I and the angle of rotation of Q about the axis of light transmission θ ?



C



D



7. [JJC/H2/Prelims 2018/P1/15]

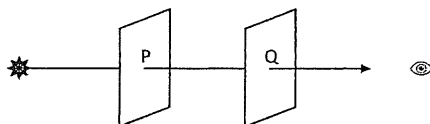
The least distance between two points of a progressive transverse wave which have a phase difference of $\frac{1}{3}\pi$ rad is 0.050 m.

If the frequency of the wave is 500 Hz, what is the speed of the wave?

- A 25 m s^{-1} B 75 m s^{-1} C 150 m s^{-1} D 1666 m s^{-1}

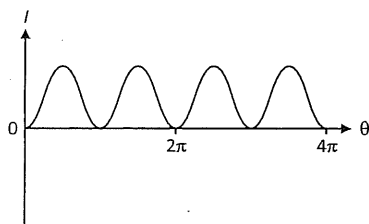
8. [JJC/H2/Prelims 2018/P1/16]

A light source is viewed through two pieces of polarizers with their axes initially at $\frac{1}{2}\pi$ rad from each other.

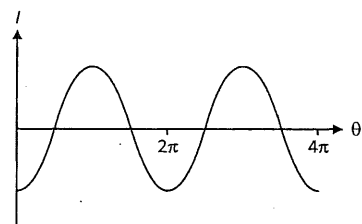


Which graph correctly shows the variation of intensity reaching the eye with angular displacement of the Q with respect to P when polarizer Q is rotated?

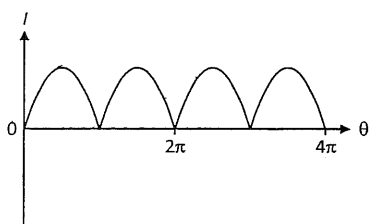
A



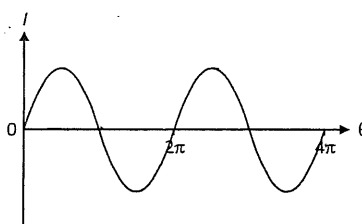
B



C

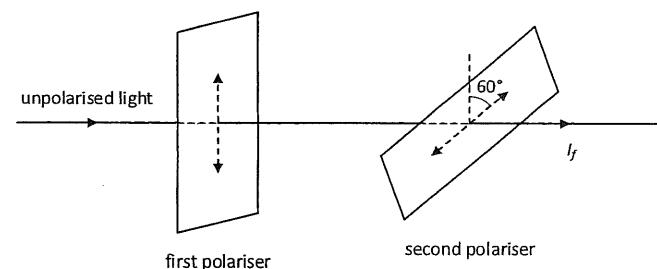


D



9. [JJC/H2/Prelims 2018/P1/20]

Unpolarised light is incident on a polariser. The light transmitted by the first polariser is then incident on a second polariser. The polarising axis of the second polariser is at 60° to that of the first polariser.



The intensity emerging from the second polariser is I_f .

Which of the following correctly gives the intensity incident on the first polariser?

- A $\frac{I_f}{8}$ B $\frac{I_f}{8}$ C $4I_f$ D $8I_f$

10. [MI/H2/Prelims 2018/P1/17]

A longitudinal wave of frequency 100 Hz is traveling in a gas at a speed of 200 m s^{-1} .

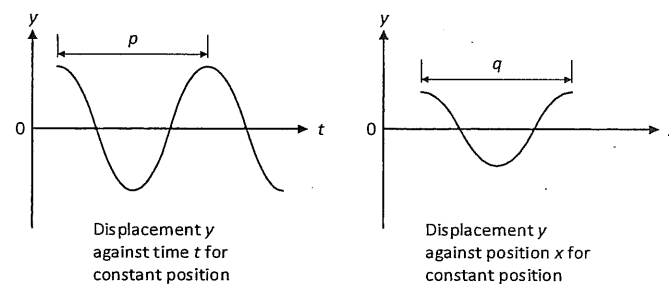
The phase difference between 2 points P and Q along the path of the wave is $\frac{5\pi}{4}$ rad.

What is the distance between points P and Q?

- A 0.525 m B 1.05 m C 1.15 m D 1.25 m

11. [MI/H2/Prelims 2018/P1/18]

The same progressive wave is represented by the following graphs.

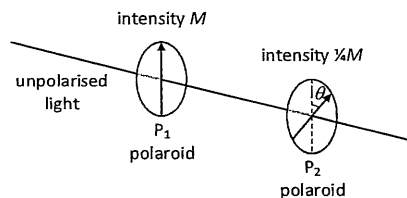


Which of the following gives the speed of propagation of the progressive wave?

- A pq B $\frac{p}{q}$ C $\frac{q}{p}$ D $\frac{1}{q}$

12. [MI/H2/Prelims 2018/P1/19]

A beam of plane-polarised light of intensity M , after passing through polaroid P_1 , falls normally on to a thin sheet of Polaroid P_2 as shown below. When polaroid P_2 is rotated through an angle θ , the transmitted beam through polaroid P_2 has an intensity of $\frac{1}{4}M$.

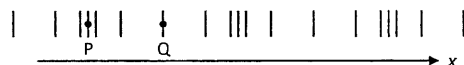


What is the angle θ between the plane of incident polarisation and the polarising direction of the polaroid P_2 ?

- A 60° B 45° C 30° D 22.5°

13. [NYJC/H2/Prelims 2018/P1/15]

The compressions and rarefactions of a sound wave travelling in air in the positive x -direction are shown below.

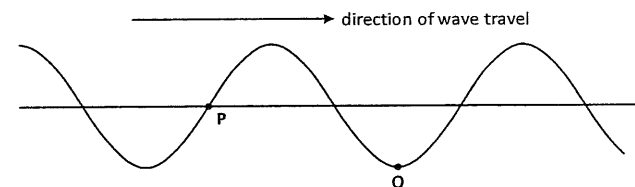


Which statement is true about the diagram?

- A At P, the air pressure is minimum.
 B At Q, it is a position of antinode.
 C At Q, the air molecule is momentarily at rest.
 D At P, the air molecule is moving in the positive x -direction.

14. [PJC/H2/Prelims 2018/P1/12]

The diagram shows a transverse wave on a rope. The wave is travelling from left to right. At the instant shown, the points P and Q on the rope have zero displacement and maximum displacement respectively.

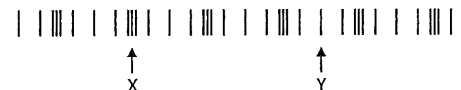


Which of the following describes the direction of motion, if any, of the points P and Q at this instant?

	P	Q
A	upwards	stationary
B	stationary	downwards
C	stationary	upwards
D	downwards	stationary

15. [PJC/H2/Prelims 2018/P1/13]

The diagram represents a longitudinal wave travelling from left to right at a frequency of 200 Hz. Two particles in the wave labelled X and Y are separated by a distance of 50 m.

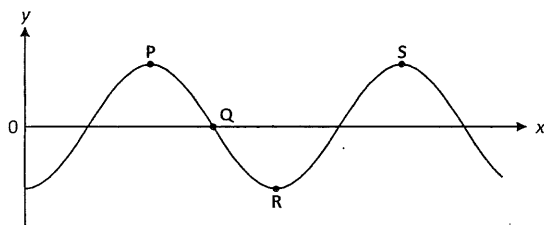


What is the velocity of the wave?

- A 2000 m s^{-1}
 B 4000 m s^{-1}
 C 6700 m s^{-1}
 D 8000 m s^{-1}

16. [SRJC/H2/Prelims 2018/P1/17]

The string shows the shape at a particular instant of part of a progressive wave travelling along a string.



Which statement about the motion of the points along the string is correct?

- A The speed at point P is maximum.
- B The displacement at point Q is always zero.
- C The energy at point R is entirely kinetic.
- D The acceleration at point S is maximum.

17. [TPJC/H2/Prelims 2018/P1/18]

A point source of sound emits energy equally in all directions at a constant rate and a person 8.0 m from the source listens. After a while, the intensity of the source is tripled.

If the person wishes the sound to seem as loud as before, how far should he be standing now?

- A 4.9 m
- B 11.5 m
- C 13.9 m
- D 24.0 m

18. [VJC/H2/Prelims 2018/P1/14]

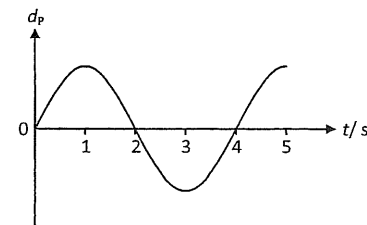
An antenna broadcasts a signal uniformly in all directions with a power of 240 W. A receiver dish of diameter 3.0 m and 750 m away receives the signal.

What is the power received by the dish?

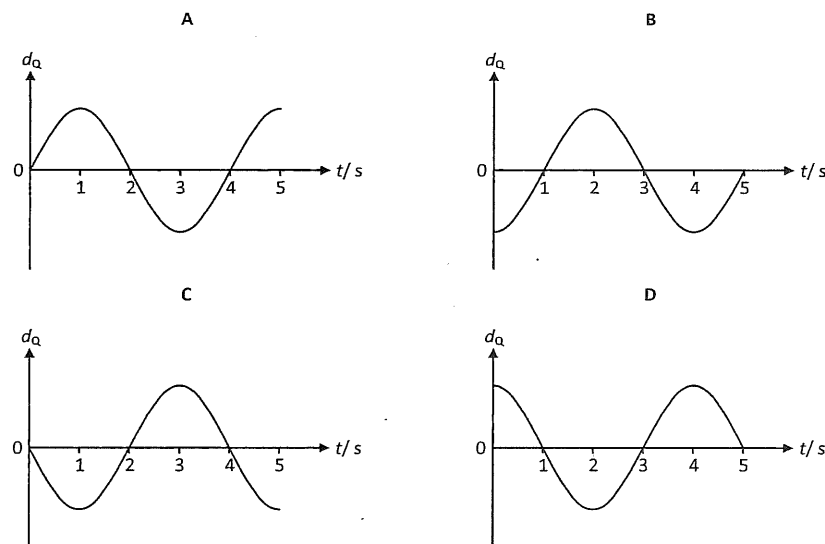
- A $9.6 \times 10^{-7} \text{ W}$
- B $3.8 \times 10^{-6} \text{ W}$
- C $2.4 \times 10^{-4} \text{ W}$
- D $2.9 \times 10^{-3} \text{ W}$

19. [VJC/H2/Prelims 2018/P1/15]

P and Q are two points 12 m apart. A wave travels from P to Q at a speed of 3.0 m s^{-1} . The graph shows the variation with time t of the displacement d_P of P.



Which of the following graphs correctly shows the variation with time t of the displacement d_Q of Q?



20. [YJC/H2/Prelims 2018/P1/15 Modified]

A source of sound of power P emits energy uniformly in all directions. A sensitive microphone, at a distance 5.0 m from the source, detects an intensity of $3.0 \times 10^{-4} \text{ W m}^{-2}$. The microphone is then moved to 10.0 m from the source.

What should the power of the source be to detect sound of intensity of $6.0 \times 10^{-4} \text{ W m}^{-2}$?

- A $2P$
- B $4P$
- C $8P$
- D $16P$

II. Structured Questions (ACJC – YJC)

1. [AJC/H2/Prelims 2018/P3/4]

The graph of a wave at time $t = 0$ is shown in Fig. 1.1. The wave is travelling to the left with a speed of 340 m s^{-1} .

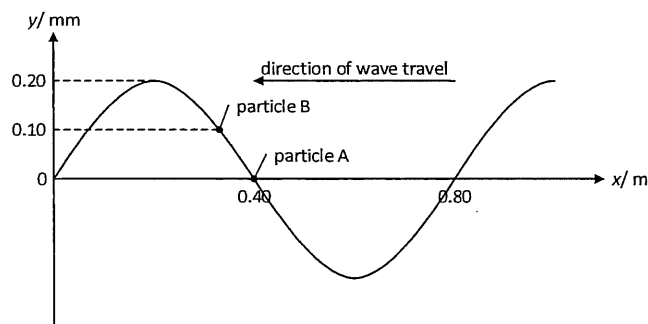


Fig. 1.1

a. Calculate the frequency of the oscillation of particle A. [2]

b. Show that the total energy E_T of the oscillation of particle A is given by

$$E_T = 2\pi^2 m f^2 a^2$$

where a is the amplitude of vibration of particle A and m is the mass of particle A. [2]

c. The mass of particle A is $4.7 \times 10^{-26} \text{ kg}$. Use the expression in (b) to calculate the energy of the oscillations of particle A. [2]

d. Sketch on the axes of Fig. 1.2 the variation with time of the displacement y for particle A. [2]

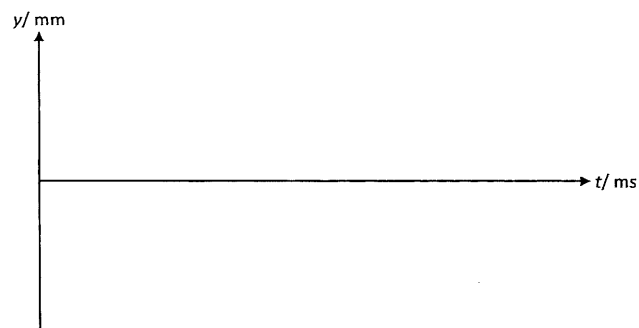


Fig. 1.2

e. Calculate the phase angle between the two particles A and B. [2]

[Total: 10]

2. [H1/H2/Prelims 2018/P2/3]

a. A commonly used quantity in astronomy is luminosity. The luminosity of a star is the total energy radiated by the star per second.

i. The luminosity of our Sun is $3.826 \times 10^{26} \text{ J s}^{-1}$. The mean distance of the Earth from the Sun is $1.496 \times 10^8 \text{ km}$. Determine the intensity of light reaching the Earth. [2]

ii. Using a photometer that detects visible light, a student measures the intensity of sunlight at noon to be less than the value calculated in a.i. Suggest a reason for this observation. [1]

iii. A photometer of area $4.00 \times 10^2 \text{ cm}^2$ is aimed directly at the sun at the top of a building. Determine the maximum power incident on the photometer. [2]

b. Fig. 2.1 shows the displacement y of a particle in a sinusoidal wave as a function of time t .

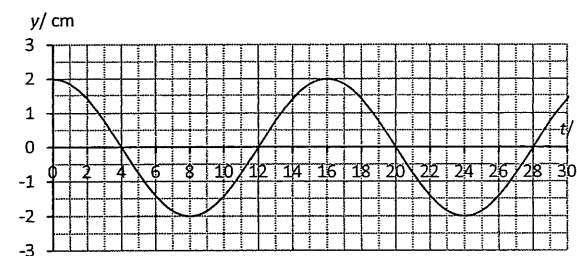


Fig. 2.1

i. Using Fig. 2.1, write an equation that represents the displacement y of the particle in terms of t . [2]

ii. A second particle is situated nearer to the source of the wave, at a distance $\frac{1}{4}\lambda$ from the first. Determine the phase difference between the vibrations of the two particles. [2]

iii. Sketch in Fig. 2.1 to illustrate the variation with time t of the displacement y of the second particle. [2]

[Total: 11]

3. [JJC/H2/Prelims 2018/P2/5 Part]

- a. Fig. 3.1 shows the variation of the height h of a water wave against its horizontal distance x at 12 noon.

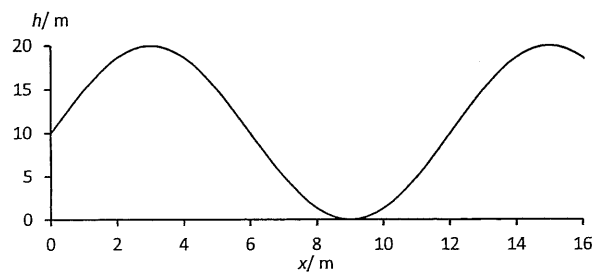


Fig. 3.1

- The waves are traveling at the speed of 1.25 m s^{-1} . Show that the frequency of the waves is 0.104 Hz . [1]
- Two buoys, fixed in position, are floating on the open sea. The waves take 2.00 minutes to travel directly from one buoy to the other buoy. Determine the phase difference of the two buoys. [2]

[Total: 3]

4. [MJC/H2/Prelims 2018/P2/2]*

A particle in a medium is oscillating because of the passage of a transverse wave W_1 . The wave has intensity I at this point. The amplitude of the oscillation is A . Fig. 4.1 shows the variation with time t of the displacement x of the particle

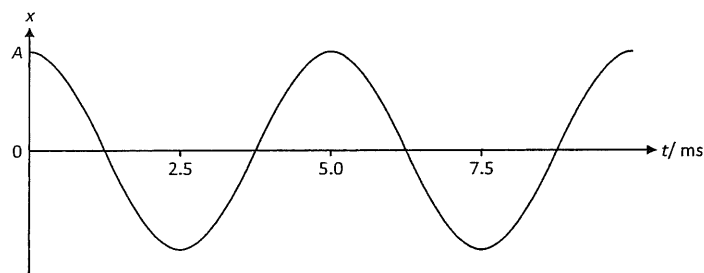


Fig. 4.1

A second, similar transverse wave W_2 has the same frequency and is incident on the same particle. The amplitude of the oscillation due to W_2 alone is $\frac{5}{2}A$ at this point.

- the frequency of the waves, [1]
 - the intensity, in terms of I , of the wave W_2 , [2]
- State two conditions which are necessary for the waves W_1 and W_2 to produce an observable interference pattern. [2]
 - State the condition that must be satisfied if the waves are to interfere to produce a minimum resultant intensity at a point. [1]
 - Calculate, in terms of I , this minimum intensity. [2]

[Total: 8]

5. [PJC/H2/Prelims 2018/P2/4]

- a. A point source of sound S emits a note of frequency 520 Hz at constant power. Two points along the wave, X and Y , separated by a distance of 0.49 m , have a phase difference of $\frac{3}{2}\pi \text{ rad}$ as shown in Fig. 5.1. Point X is 5.24 m away from S .

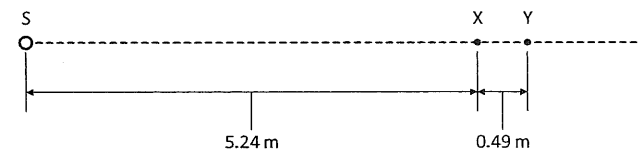


Fig. 5.1

- Determine the wavelength and speed of the sound wave. [3]
 - Given that the intensity at X is $2.6 \times 10^{-3} \text{ W m}^{-2}$, calculate the intensity of the sound at Y . [2]
- b. A parallel beam of light is incident on two polarisers P and Q . When the polarising directions are parallel, the amplitude of the emergent beam is A as shown in Fig. 5.2.

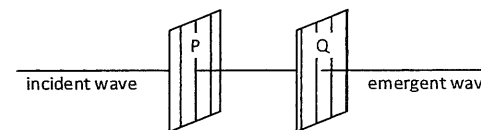


Fig. 5.2

- Determine the angle through which Q must be rotated so that the amplitude of the emergent beam is reduced to $\frac{1}{2}A$. [2]
- Determine the corresponding fractional change in the intensity of the emergent beam. [2]

[Total: 9]

6. [TPJC/H2/Prelims 2018/P2/4 Modified]

- a. With reference to the vibrations of the particles on a wave and its direction of energy transfer, distinguish between transverse waves and longitudinal waves. [2]
- b. Fig. 6.1 shows the equilibrium positions of the particles in a medium when they are undisturbed as compared to their displaced positions in the presence of a progressive sinusoidal wave. Displacement to the right is taken as positive.

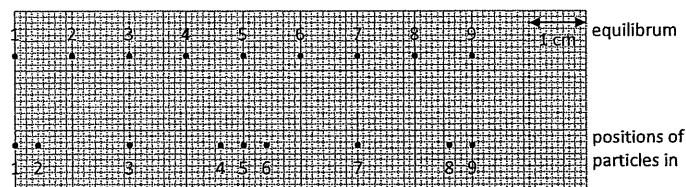


Fig. 6.1

- i. Deduce the displacement of particle 2 when the wave passes through it at the instant shown in Fig. 6.1. [1]
- ii. Sketch, in Fig. 6.2, the displacement-position graph representing all the particles disturbed by the wave as shown in Fig. 6.1. Indicate on your graph all centres of rarefaction R and compression C. [3]

displacement/ cm

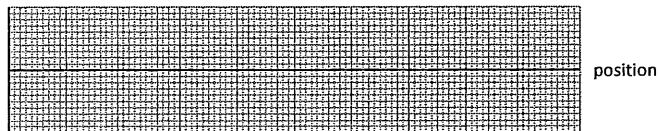


Fig. 6.2

- iii. Hence deduce the amplitude of the progressive wave. [1]
- iv. State the phase difference between particles 5 and 7. [1]

[Total: 8]

